Daniel E. Hernández Lang, Ph.D.

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Education

2020 – 2025 **Ph.D. in Astronomy**, Ludwig-Maximilians-Universität München, Germany.
Thesis title: *From galaxy clusters to cosmology.*

2016 – 2019 **Master in Astronomy**, Univerisdad de La Serena in High Energy Magic.
Thesis title: *Morphology of galaxies in low-mass galaxy clusters and its relation with matter distribution.*

2012 – 2016 **Bachelor in Phsycis, Astronomy specialization**, Universidad de La Serena, Chile.

Experience

Research

2025 – · · · · **Postdoctoral researcher**, LMU, Munich.
Project: *Euclid*

2024 – 2025 **Postdoctoral researcher**, LMU, Munich.
Project: *PUNCH4NFDI*

2023 – 2024 **Postdoctoral researcher**, LMU, Munich.
Project: *MeerKLASS*

2020 – 2025 **PhD**, LMU, Munich.
Supervisor: *Prof. Dr. Joseph Mohr*

(04 – 07)/2019 **Research assistant**, AURA, La Serena.
Supervisor: *Dr. Rodrigo Carrasco*.

2016 - 2019 **Master,**, Universidad de La Serena, La Serena.
Supervisor: *Dr. Sergio Torres Flores*.

2017 - 2018 **Research assistant**, AURA, La Serena.
Supervisor: *Dr. Alfredo Zenteno*.

08/2018 Utilization of supervised learning for the classification of lymphomas, La Serena School for Data Science, La Serena.

Observational runs & Proposals

2022 **Co-I** on accepted proposal with the Hubble Space Telescope, STScI, *"Clash of Titans: Characterizing SPT-CL J0307-6225, a major merger in the plane of the sky"*.
PI: *Zenteno, A.*.

Co-I on accepted proposal with the GMOS-S MOS instrument, Gemini, *"A dynamical study of massive galaxy clusters at z>0.6 undergoing extreme mergers"*
PI: *Carrasco, R.*. IDs: GS-2022B-Q-317, GS-2022B-Q-127.

2019 – 2021 **Co-I** on accepted proposals with the DECam instrument at the BLANCO telescope, CTIO, *"The DECam eROSITA Survey (DeROSITAS)"*
PI: *Zenteno, A.*. IDs: 2019A-0272, 2019B-0323, 2020A-0399, 2020B-0241, 2021A-0149.

2021 **Co-I** on accepted proposal with the GMOS-S MOS instrument, Gemini, *"Witnessing the formation of the large-scale structure: a dynamical study of massive galaxy clusters undergoing extreme mergers"*
PI: *Monteiro-Oliveira, R.*. ID: GS-2021B-Q-302.

Experience (continued)

2020 — **Co-I** on accepted proposal with the GMOS-S MOS instrument, Gemini, *"Clash of Titans: a dynamical study of a sample of massive galaxy clusters undergoing extreme mergers"* PI: *Zenteno, A.*. IDs: GS-2020B-Q-235, GS-2020B-Q-131.

2018 — **Co-I** on accepted proposal with the GMOS-S MOS instrument, Gemini, *"Clash of Titans: a dynamical study of a sample of massive galaxy clusters undergoing extreme mergers"* PI: *Zenteno, A.*. ID: GS2018B-Q-233.

Large Collaborations

2025 – · · · · — **Member and developer** of the Euclid collaboration.

2021 – · · · · — **Builder and observer** of the DECam Local Volume Exploration survey.

2018 – · · · · — **Member** of the Dark Energy Survey collaboration.

2024 – 2025 — **Member and tester** of the PUNCH4NFDI consortium.

2019 – 2021 — **Member** of the eROSITA-DE collaboration.

Others

(01 – 04)/2017 — **In charge** of operations and software development at the FALCON-MMO Telescope.

(03 – 12)/2016 — **Coordinator** of outreach activities of the FALCON-MMO project.

Research Publications

First Author

1. **D. Hernández-Lang** et al., *The PSZ-MCMF catalogue of Planck clusters over the DES region*, Oct. 2023. DOI: 10.1093/mnras/stad2319. arXiv: 2210.04666 [astro-ph.CO].
2. **D. Hernández-Lang** et al., *Clash of Titans: A MUSE dynamical study of the extreme cluster merger SPT-CL J0307-6225*, Dec. 2022. DOI: 10.1093/mnras/stac2480. arXiv: 2111.15443 [astro-ph.GA].

Contributing Author

1. A. Zenteno, M. Kluge, R. Kharkrang, **D. Hernandez-Lang**, et al., *The dynamical state of eROSITA clusters and its impact on the brightest cluster galaxy luminosity*, Jun. 2025. DOI: 10.1051/0004-6361/202452440. arXiv: 2503.21066 [astro-ph.CO].
2. C. Lima-Dias, A. Monachesi, S. Torres-Flores, A. Cortesi, **D. Hernández-Lang**, et al., *Bulge-disc decomposition of the Hydra cluster galaxies in 12 bands*, Jan. 2024. DOI: 10.1093/mnras/stad3571. arXiv: 2311.09313 [astro-ph.GA].
3. F. Aldás, A. Zenteno, F. A. Gómez, **D. Hernández-Lang**, et al., *Clash of Titans: the impact of cluster mergers in the galaxy cluster red sequence*, Oct. 2023. DOI: 10.1093/mnras/stad2261. arXiv: 2307.11837 [astro-ph.GA].
4. M. Klein, **D. Hernández-Lang**, et al., *RASS-MCMF: a full-sky X-ray selected galaxy cluster catalogue*, Dec. 2023. DOI: 10.1093/mnras/stad2729. arXiv: 2305.20066 [astro-ph.CO].
5. C. Lima-Dias, A. Monachesi, S. Torres-Flores, A. Cortesi, **D. Hernández-Lang**, et al., *An environmental dependence of the physical and structural properties in the Hydra cluster galaxies*, Jan. 2021. DOI: 10.1093/mnras/staa3326. arXiv: 2010.15235 [astro-ph.GA].
6. A. Zenteno, **D. Hernández-Lang**, et al., *A joint SZ-X-ray-optical analysis of the dynamical state of 288 massive galaxy clusters*, Jun. 2020. DOI: 10.1093/mnras/staa1157. arXiv: 2004.01721 [astro-ph.GA].
7. E. J. Gonzalez, M. de los Rios, G. A. Oio, **D. Hernández-Lang**, et al., *Analysis of candidates for interacting galaxy clusters. I. A1204 and A2029/A2033*, Mar. 2018. DOI: 10.1051/0004-6361/201732003. arXiv: 1801.01498 [astro-ph.CO].

Nth Author

1. M. Salvato et al., *Counterpart identification and classification for eRASS1 and characterisation of the active galactic nuclei content*, Dec. 2025. DOI: 10.1051/0004-6361/202556142. arXiv: 2509.02842 [astro-ph.HE].
2. M. Kluge et al., *The SRG/eROSITA All-Sky Survey. Optical identification and properties of galaxy clusters and groups in the western galactic hemisphere*, Aug. 2024. DOI: 10.1051/0004-6361/202349031. arXiv: 2402.08453 [astro-ph.CO].
3. W. Cerny et al., *Pegasus IV: Discovery and Spectroscopic Confirmation of an Ultra-faint Dwarf Galaxy in the Constellation Pegasus*, Jan. 2023. DOI: 10.3847/1538-4357/aca1c3. arXiv: 2203.11788 [astro-ph.GA].
4. Euclid Collaboration et al., *Euclid preparation. XXV. The Euclid Morphology Challenge: Towards model-fitting photometry for billions of galaxies*, Mar. 2023. DOI: 10.1051/0004-6361/202245041. arXiv: 2209.12906 [astro-ph.GA].
5. Euclid Collaboration et al., *Euclid preparation. XXVI. The Euclid Morphology Challenge: Towards structural parameters for billions of galaxies*, Mar. 2023. DOI: 10.1051/0004-6361/202245042. arXiv: 2209.12907 [astro-ph.GA].
6. A. Drlica-Wagner et al., *The DECam Local Volume Exploration Survey Data Release 2*, Aug. 2022. DOI: 10.3847/1538-4365/ac78eb. arXiv: 2203.16565 [astro-ph.IM].
7. W. Cerny et al., *Eridanus IV: an Ultra-faint Dwarf Galaxy Candidate Discovered in the DECam Local Volume Exploration Survey*, Oct. 2021. DOI: 10.3847/2041-8213/ac2d9a. arXiv: 2107.09080 [astro-ph.GA].
8. A. Drlica-Wagner et al., *The DECam Local Volume Exploration Survey: Overview and First Data Release*, Sep. 2021. DOI: 10.3847/1538-4365/ac079d. arXiv: 2103.07476 [astro-ph.GA].
9. T. H. Reiprich et al., *The Abell 3391/95 galaxy cluster system. A 15 Mpc intergalactic medium emission filament, a warm gas bridge, infalling matter clumps, and (re-) accelerated plasma discovered by combining SRG/eROSITA data with ASKAP/EMU and DECam data*, Mar. 2021. DOI: 10.1051/0004-6361/202039590. arXiv: 2012.08491 [astro-ph.CO].
10. S. Mau et al., *Two Ultra-faint Milky Way Stellar Systems Discovered in Early Data from the DECam Local Volume Exploration Survey*, Feb. 2020. DOI: 10.3847/1538-4357/ab6c67. arXiv: 1912.03301 [astro-ph.GA].

Conference Proceedings

1. **D. Hernández-Lang** et al., "A MUSE dynamical study of the cluster merger SPT-CLJ 0307-6225," in *Galaxy Cluster Formation II*, Jun. 2021, 66, p. 66. DOI: 10.5281/zenodo.5031392.

Skills

Languages	**Spanish** (native), **English** (proficient), and **German** (basic).
Coding	Python, SQL, XML/XSL, LaTeX, bash, and git.
Softwares	SExtractor, SourceExtractor++, PSFex, IRAF, GALFIT, TOPCAT, GMMPS, THELI, Aladin, DS9.
Operative systems	Windows, Linux, and MacOS.